EXECUTIVE SUMMARY

Portfolio Theory in Practice

FE5108 Portfolio Theory and Investments | Track A
National University of Singapore | 15 October 2026

Decision for the investment committee

Use value weights as a benchmark, but permit limited active deviations under concentration, leverage and short-sale limits. The unconstrained tangency solution is informative about the historical opportunity set, yet too dependent on estimated inputs to serve as a literal implementation mandate.

What the evidence says

Stage	Principal finding	Investment implication
1 | Portfolio choice	In-sample annualized Sharpe: 2.197 versus 1.269 for value weighting. Tangency gross exposure: 5.389 times capital.	The optimizer improves its fitted objective by taking large offsetting positions; this is not a forecast.
2 | CAPM	GRS F(30,77) = 1.1465, p = 0.3099. MSFT and WMT cross the individual |t| > 2 screen.	Individual signals coexist with joint non-rejection; neither finding proves or disproves the CAPM on its own.
3 | FF3	Absolute alpha falls for 13/30 stocks. MSFT retains its alpha screen; WMT does not.	Size and value exposures matter, but do not explain away all estimated abnormal returns.
4 | Judgment	With MSCI World, GRS p = 0.1154; AAPL joins the individually significant stocks.	The proxy affects estimates and stock-level judgments, while joint non-rejection is unchanged.

The main evidence uses **30 US stocks and 108 monthly observations, January 2017-December 2025**. One eligible capitalization leader is selected from each of 30 text-based classes. Monthly total returns include distributions; the common risk-free series and original market proxy come from the cached Kenneth French factors.

The important qualification is retrospective design. End-2024 classifications, complete-history eligibility and end-2025 value weights are applied to earlier observations. A supplementary January-October 2026 tracking exercise supports continued investigation of optimization, but is not a clean prospective test and excludes full implementation costs.

Least trusted inference: treating the fitted Sharpe of 2.197 as future achievable performance. Shrinking the covariance estimate and imposing implementable constraints are more defensible than taking the unrestricted weights at face value.

Report guide: data and design, p. 2; Stage 1, pp. 3-4; Stage 2, pp. 5-6; Stage 3, pp. 7-8; investment-committee memo (Stage 4), pp. 9-10. Supporting tables, figures, sources and the contribution statement follow in the appendix.

RESEARCH DESIGN

Data, universe and measurement

A diversified research menu, not the entire market

The starting partition groups 1,910 firms into 30 classes using the retained end-2024 annual-report text representation (TF-IDF/SVD, UMAP3 and Leiden; seed 20260831). Within each class, we select the largest company by the frozen December 31, 2024 vendor capitalization that has complete 2017-2025 returns. This preserves all 30 classes rather than selecting stocks solely by realized performance. Twenty-three selections are absolute class capitalization leaders; seven are coverage substitutes. Text-class membership does not guarantee return diversification.

Design choice	Implementation and rationale
Universe and frequency	30 securities; 108 monthly observations each (3,240 stock-months). The main sample exceeds the 96-month course minimum.
Stock returns	Processed TBIC panels using Massive prices and corporate actions. Use annual evaluation windows only to avoid duplicating overlapping historical observations.
Monthly aggregation	R(i,m) = product over days of [1 + r(i,d)] - 1. Require finite returns on every panel trading date; no zero-filling or price carrying.
Risk-free proxy	Monthly US one-month Treasury-bill RF in the cached French file, in USD and decimal units. Subtract once after monthly compounding.
Original market proxy	French US Mkt-RF: a broad US equity excess-return proxy aligned with the stocks' currency and market. It is not the theoretical all-asset portfolio.
Value benchmark	Fixed December 31, 2025 company-capitalization weights within the same 30 stocks; not a float-adjusted broad-market index or historical monthly cap-weighted strategy.
Alternative proxy	MSCI World Standard (large and mid cap), USD net total return, index 990100. Monthly index return minus the same RF.

Timing, selection and source limitations

The 2024 classification and full-history eligibility are applied retrospectively. Excluding incomplete histories may favor survivors and affect means, covariances and inferred performance; the direction of every individual beta or alpha distortion is not determined. The project does not claim the selected basket was investable under these rules in 2017. The seven substitutions reflect panel coverage, not proof that the omitted firms did not trade earlier.

Upstream processing accounts for splits, cash distributions and verified terminal events where available. Identity and internal arithmetic checks do not independently certify every vendor record or historical share count. Selection uses 2024 capitalization; valuation weights use the distinct 2025 snapshot. Both have stated share-timing limitations.

The repository reproduces downstream analyses from cached inputs. Rebuilding monthly returns requires excluded local daily panels; it does not reconstruct the entire vendor data lake. Appendix B explains the reproduction boundary.

Sources: data documentation; universe selection; monthly-return and MSCI definitions.

STAGE 1 | PORTFOLIO CHOICE

Estimated moments and the risky portfolio

Let μ be the vector of arithmetic monthly mean total returns, Σ the sample covariance of total returns (divisor T - 1), and μ_e = μ - mean(RF)·1. The project solves Σz = μ_e and normalizes **w = z / (1′z)**; shorts are allowed and net risky weights sum to one. This is the textbook tangency solution under the estimated moments. With a common risk-free asset and the model's assumptions, investors would vary their exposure to this risky fund with risk tolerance; the empirical challenge is estimating that fund reliably.

Table 1. Universe summary statistics. All figures below except observation counts are monthly percentages. The means are arithmetic, not compound growth rates.

Stock	Months	Mean total (%)	Mean excess (%)	Volatility (%)
SHW	108	1.55	1.36	7.52
AAPL	108	2.49	2.29	8.02
ABBV	108	1.83	1.64	7.46
PLD	108	1.32	1.13	7.29
TMO	108	1.57	1.38	6.86
SF	108	1.71	1.52	8.47
VZ	108	0.33	0.13	5.15
META	108	2.19	2.00	10.51
SCCO	108	2.25	2.06	10.01
MSFT	108	2.19	1.99	5.89
UPS	108	0.51	0.32	8.22
WMT	108	1.76	1.57	5.50
PANW	108	2.56	2.37	10.31
RTX	108	0.95	0.76	8.14
AFL	108	1.47	1.28	6.35
ACN	108	1.14	0.95	6.86
V	108	1.62	1.43	5.80
EXEL	108	1.68	1.49	12.02
KO	108	0.86	0.67	4.73
UNH	108	1.11	0.92	7.76
WMB	108	1.39	1.20	8.12
MO	108	0.67	0.48	6.71
AGNC	108	0.90	0.71	8.21
CME	108	1.24	1.05	5.33
SRE	108	0.95	0.76	5.77
MAR	108	1.77	1.58	9.58
CMG	108	1.97	1.78	10.07
TSLA	108	4.86	4.67	18.97
SLB	108	0.31	0.12	12.47
DHI	108	2.20	2.01	10.60

Dispersion is substantial: TSLA has the highest average monthly total return (4.86%) and volatility (18.97%), while KO has the lowest volatility (4.73%). High standalone mean return does not by itself determine the tangency weight: covariance with the rest of the basket matters.

Source: artifacts/stage1/summary_statistics.csv.

STAGE 1 | PORTFOLIO CHOICE

Weights reveal the cost of fitted efficiency

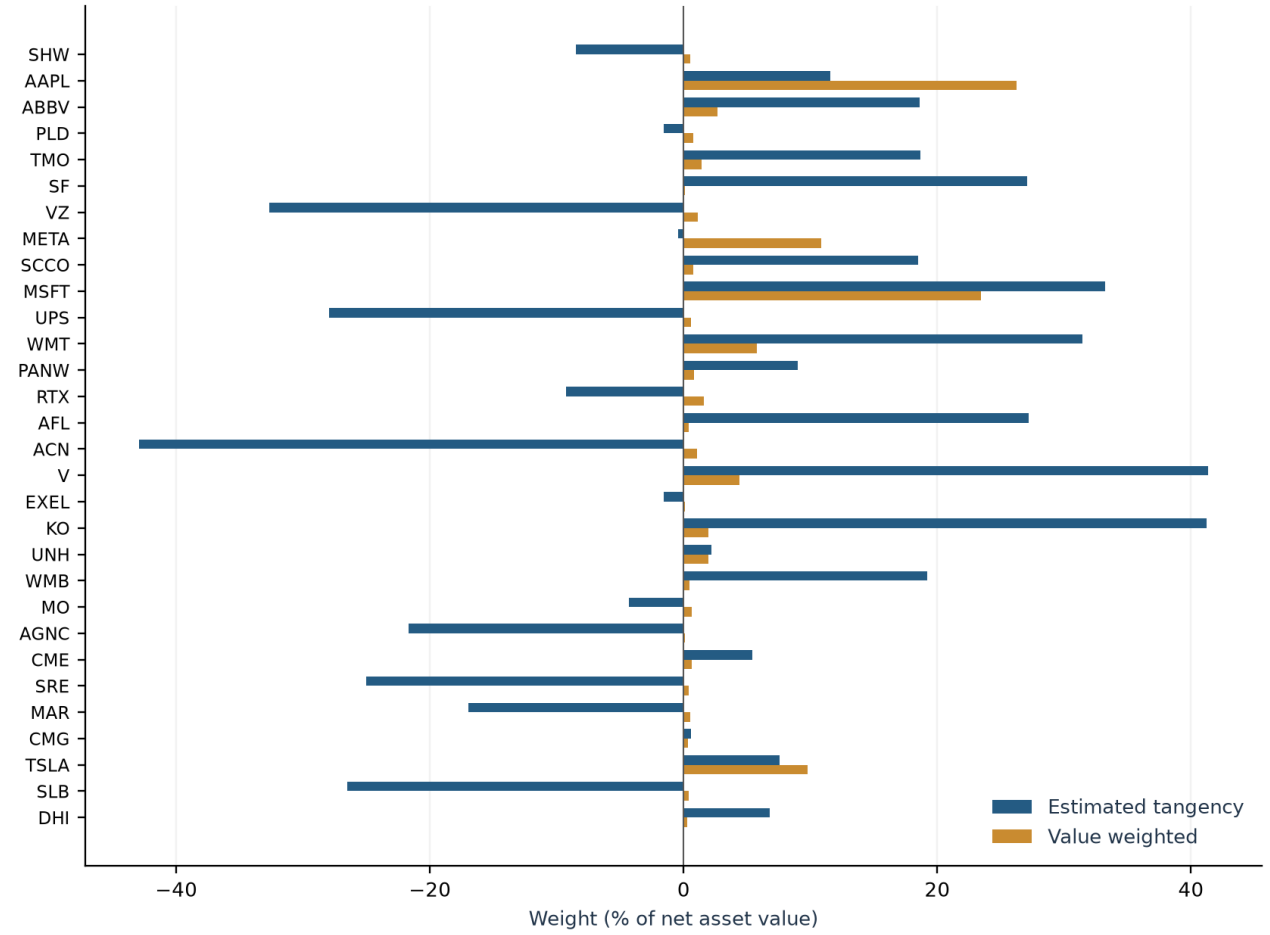


Figure 1. Tangency versus value weights, 108-month sample. Takeaway: similar net investment conceals radically different gross exposure; the optimized portfolio relies on large long and short positions. Value weights use the fixed December 2025 snapshot. Replotted from portfolio_weights.csv.

Estimation window	Tangency Sharpe	Gross exposure	Mean |Δw| (pp)	Correlation with 108m
108 months	2.197	5.389×	0.00	1.000
96 months	2.077	6.122×	5.85	0.965
84 months	2.484	8.812×	17.08	0.848

The full-sample tangency portfolio has about 319.45% long and 219.45% short exposure; individual weights range from -42.96% (ACN) to +41.36% (V). Its fitted annualized excess mean, volatility and Sharpe are 42.81%, 19.48% and 2.197, versus 25.37%, 19.99% and 1.269 for value weighting. Annualization uses 12 times the monthly mean and √12 times volatility; these are fitted moment summaries, not realized compound returns.

Dropping the earliest 12 or 24 months changes weights even with the same endpoint and universe. Mean noise and inversion of covariance can magnify estimation error; economic regime changes may also contribute. The 84-month exercise is a supplemental sensitivity check, not the main sample satisfying the eight-year requirement. A higher fitted Sharpe in that window does not establish better future performance.

Sources: weights; portfolio diagnostics; sensitivity measures. Full weights and sensitivity figure: Appendices C1-C3.

STAGE 2 | CAPM

Security characteristic regressions

For each stock, estimate $R_{i,t} - RF_t = \alpha_i + \beta_i(R_{M,t} - RF_t) + \varepsilon_{i,t}$ using the French Mkt-RF factor and an intercept. β measures market co-movement; α is the average return left unexplained by that proxy. All 30 regressions use the same 108 months.

Table 2. CAPM estimates. Alpha is percent per month; t(α) uses ordinary OLS standard errors with 106 residual degrees of freedom. R^2 is not adjusted. Full OLS and HAC(3) inference is in Appendix D.

Stock	Beta	Alpha (%/mo)	t(alpha), OLS	R^2
SHW	1.087	0.191	0.35	0.456
AAPL	1.125	1.082	1.80	0.429
ABBV	0.535	1.065	1.52	0.112
PLD	1.014	0.039	0.07	0.422
TMO	0.823	0.492	0.87	0.312
SF	1.287	0.130	0.22	0.504
VZ	0.337	-0.229	-0.47	0.094
META	1.208	0.699	0.80	0.290
SCCO	1.089	0.890	1.04	0.259
MSFT	0.894	1.032	2.50	0.501
UPS	1.014	-0.775	-1.16	0.330
WMT	0.524	1.008	2.07	0.199
PANW	1.125	1.159	1.32	0.260
RTX	0.685	0.018	0.02	0.156
AFL	0.785	0.430	0.84	0.334
ACN	1.117	-0.254	-0.57	0.576
V	0.853	0.515	1.23	0.472
EXEL	0.654	0.787	0.68	0.065
KO	0.434	0.201	0.47	0.184
UNH	0.548	0.330	0.45	0.108
WMB	1.032	0.084	0.13	0.353
MO	0.496	-0.058	-0.09	0.120
AGNC	1.178	-0.558	-0.93	0.451
CME	0.324	0.698	1.38	0.081
SRE	0.647	0.065	0.13	0.274
MAR	1.473	-0.009	-0.01	0.517
CMG	1.126	0.567	0.66	0.272
TSLA	2.018	2.491	1.52	0.247
SLB	1.551	-1.550	-1.54	0.337
DHI	1.488	0.408	0.51	0.430

MSFT (t = 2.50) and WMT (t = 2.07) cross the descriptive |t| > 2 screen. A large estimate need not be precise: TSLA's alpha is about 2.49% per month but its t-statistic is only 1.52. The screen is unadjusted for multiple comparisons and is not a declaration of mispricing.

HAC inference uses three Bartlett lags, a small-sample correction and an approximate Student-t reference. It is reported separately from OLS; it is not substituted into the classical GRS formula.

Source: artifacts/stage2/capm_results.csv.

STAGE 2 | CAPM

The SML and the joint alpha assessment

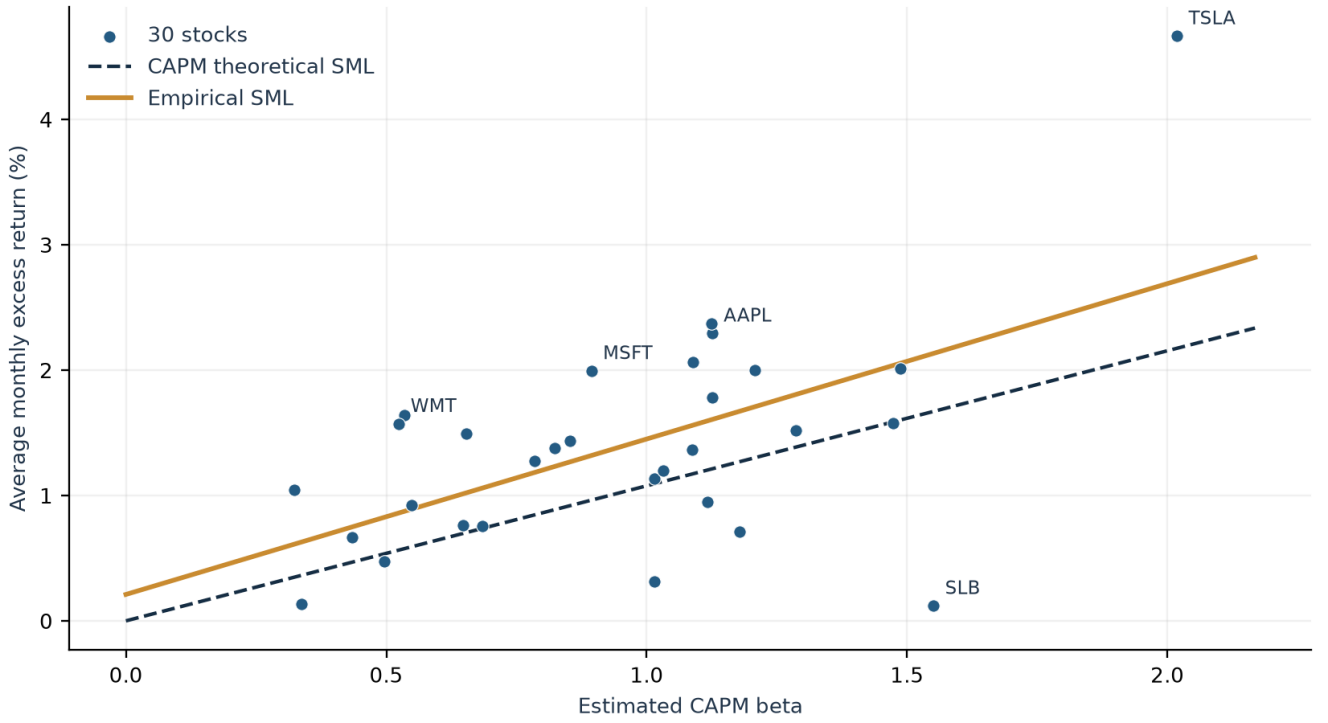


Figure 2. Empirical and theoretical Security Market Lines. Takeaway: the fitted line has a positive intercept and is steeper, not flatter, than the theoretical line in this selected sample. The vertical axis is excess return, so the theoretical line passes through zero. Replotted from Stage 2 sml_points.csv and sml_lines.csv.

Monthly SML quantity	Theory	Estimated	Difference (pp)
Intercept	0.0000%	0.2108%	+0.2108
Slope per unit beta	1.0771%	1.2396%	+0.1624

The empirical line is an equal-weight cross-sectional regression of mean stock excess returns on estimated betas ($R^2 = 0.3164$). Its intercept is not an individual time-series alpha. These slope/intercept differences are descriptive: estimated regressors and cross-stock dependence prevent treating ordinary cross-sectional p-values as a formal pricing test. A nonzero intercept at beta zero may also involve extrapolation.

Joint inference is different from screening individual stocks

Under H_0: $\alpha_1 = \ldots = \alpha_{30} = 0$, the classical GRS statistic is $[(T - N - K)/N] \times [\alpha' \Sigma_\varepsilon^{-1} \alpha] / [1 + \text{mean}(f)^2/\text{var}(f)]$. Here $T = 108$, $N = 30$, $K = 1$; $\Sigma_\varepsilon = E'E/T$ and market variance uses divisor T. Its finite-sample reference is F(30,77), under the normality, constant nonsingular covariance and time-independence assumptions of Gibbons, Ross and Shanken (1989).

GRS statistic	p-value	5% critical value	Decision
1.1465	0.3099	1.6073	Do not reject joint zero alpha

At a nominal 5% size, 30 valid individual tests would generate about 1.5 rejections on average under all-zero alphas, even without assuming independence for that expectation. Two screened stocks are therefore weak evidence by themselves. Joint non-rejection does not prove that all alphas are zero or that every individual alpha is insignificant.

Sources: SML summary; GRS results.

STAGE 3 | FACTOR MODELS

Adding size and value exposures

We augment the market regression with two published factors, following Fama and French (1993): $R_{i,t} - RF_t = \alpha_i + \beta_{M,i}Mkt\text{-}RF_t + \beta_{S,i}SMB_t + \beta_{H,i}HML_t + \varepsilon_{i,t}$.
The stock sample and dates are unchanged. We use cached published factors rather than constructing factors from the 30 selected stocks.

Factor construction and interpretation

SMB (Small Minus Big). The published three-factor SMB is the average return on three small-company size/book-to-market portfolios minus the average return on the three corresponding big-company portfolios. The constituent portfolios use prior accounting and market-capitalization information. SMB is a return spread, not an individual company's market capitalization. A negative loading indicates co-movement toward the big-company side of this factor.

HML (High Minus Low). The published HML is the average return on the small- and big-company high book-to-market portfolios minus the average on their low book-to-market counterparts. It is a value-minus-growth return spread, not the book-to-market ratio of a stock. A negative loading indicates exposure toward the low book-to-market side. Both factor columns already use monthly decimal returns; RF is not subtracted from these spreads again.

What happens to the Stage 2 alpha screens?

Stock	CAPM alpha (%/mo), t	FF3 alpha (%/mo), t	SMB beta	HML beta	Joint loading p
MSFT	1.032, 2.50	0.693, 2.07	-0.473	-0.545	2.94×10^{-11}
WMT	1.008, 2.07	0.737, 1.56	-0.528	-0.141	0.00455

MSFT's alpha shrinks but remains above the |OLS t| > 2 screen. WMT's alpha also shrinks but no longer crosses it. The 5% inference is consistent under HAC(3): MSFT p = 0.0454; WMT p = 0.0659. Thus, WMT's Stage 2 signal does not survive this specification at 5%, whereas MSFT's does. This is evidence of model sensitivity, not proof that WMT was mispriced or that all of its abnormal return has been economically explained.

The joint SMB/HML p-value tests whether the two **factor loadings** are jointly zero for a stock. It is neither a test that alpha is zero nor an FF3 GRS test. A significant loading or a higher fit is not enough to establish a priced premium or profitable trading strategy.

Sources: factor definitions; factor interpretation; full estimates.

STAGE 3 | FACTOR MODELS

The factor adjustment is not universal

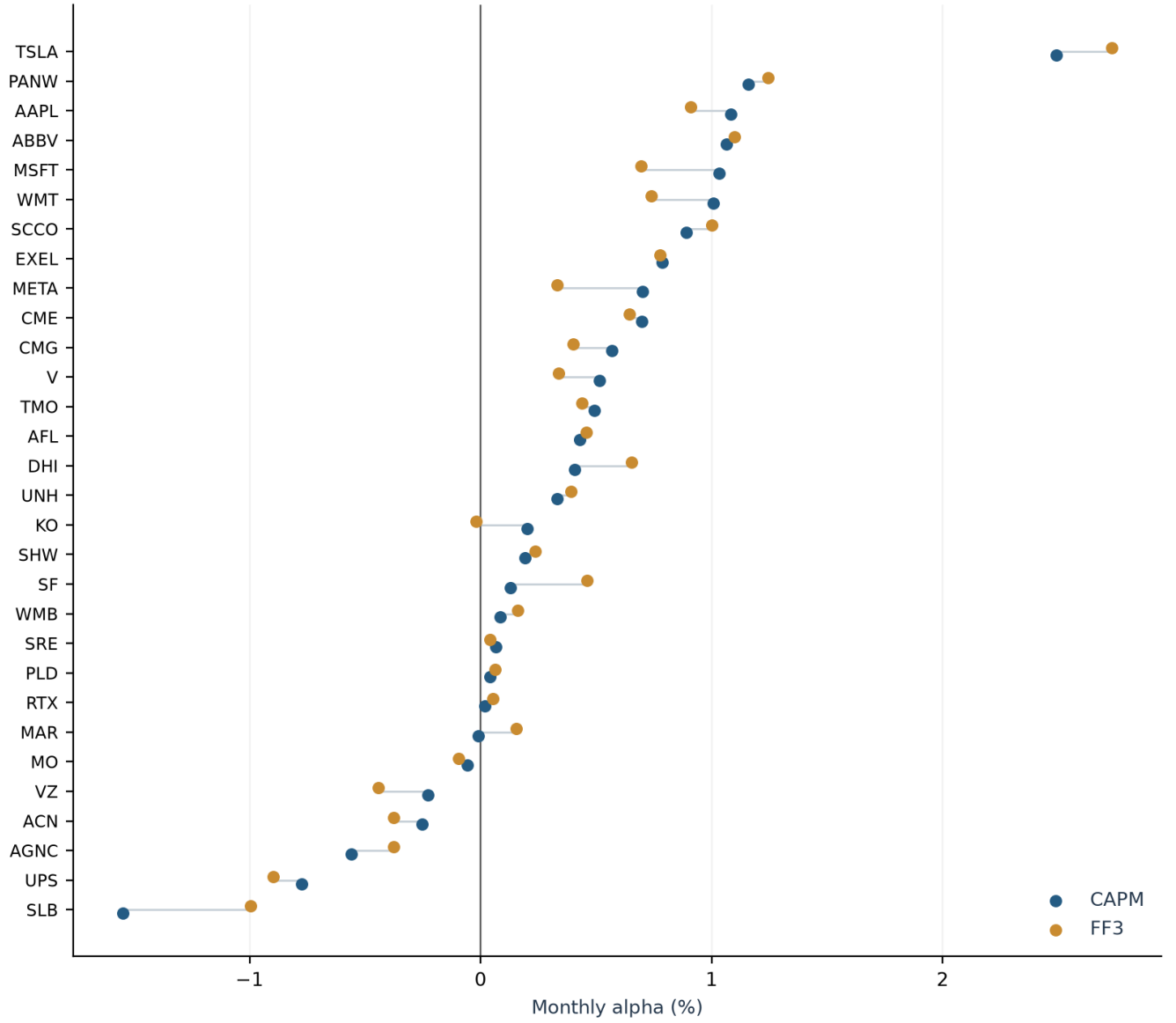


Figure 3. CAPM versus FF3 monthly alphas, all 30 stocks. Takeaway: the two initially screened alphas shrink, but only 13 of all 30 absolute alphas decline. Adding factors does not uniformly eliminate estimated abnormal returns. Dots are point estimates, not confidence intervals; connecting segments identify the same stock.

Across the universe, mean absolute alpha falls only modestly, from **0.594% to 0.574% per month**. Mean adjusted R^2 rises from 0.298 to 0.364, and 19/30 stocks have a joint SMB/HML loading-test p-value below 0.05. These descriptive aggregates were calculated from the saved estimates; the 30 loading tests are not adjusted for multiplicity.

The combination of improved fit and persistent alphas cautions against a binary verdict. Additional factors capture meaningful co-movement, yet omitted exposures, sampling uncertainty and retrospective selection remain possible explanations. MSFT's surviving signal is a candidate for further investigation, not an established arbitrage.

Source: Stage 3 full estimates. Replotted comparison; the repository counterpart is alpha_comparison.png. Full loadings and inference appear in Appendix E.

STAGE 4 | INVESTMENT-COMMITTEE MEMO (1 OF 2)

Recommendation: constrain the active allocation

To the investment committee: choose an allocation between the fixed value-weighted basket and the unconstrained tangency portfolio. Use value weights as the benchmark, control concentration, and permit limited active deviations informed by more stable covariance estimates. Set individual-position, gross-exposure and short-selling limits before implementation. The present evidence does not identify an optimal blending percentage.

Stage 1's higher fitted Sharpe favors studying optimization, but its 5.389-times gross exposure and sensitivity to the estimation window argue against literal implementation. Stage 2 does not reject joint zero alpha, while Stage 3 shows that some individual signals weaken after adding size and value. These are asset-pricing tests, not direct forecasts of the optimizer's trading performance.

Supplementary evidence, not an additional course stage

2026 tracking allocation	Return	Max. drawdown	Initial gross
Original tangency	40.78%	-23.70%	5.389×
Ledoit-Wolf tangency	31.30%	-16.83%	3.494×
Value weighted	11.33%	-13.09%	1.000×
Same-stock equal weight	11.93%	-8.40%	1.000×

Table 3. January 2, 2026 open to October 6 close; cumulative, not annualized, returns. Initial shares are held with corporate-action adjustments and retained cash dividends, rather than periodic weight rebalancing. Tracking is outside the weight-estimation window but retrospectively designed.

The original tangency allocation clearly leads over this period, so it should not be dismissed merely because its in-sample objective was optimized. However, its drawdown is deeper, and the comparison excludes full financing, borrow availability, trading costs, taxes and margin liquidation. Portfolios are not risk matched. The 2026 results therefore do not establish superior achievable risk-adjusted returns.

Ledoit-Wolf changes covariance estimation while leaving the estimated excess-mean vector unchanged. Lower gross exposure and a smaller drawdown with substantial retained return support more stable optimization. Some of that improvement may simply reflect less exposure; the remaining 3.494-times gross position is still aggressive. This is a robustness comparison, not proof of a universally superior allocation.

Style uncertainty reinforces the case for diversification

The value basket itself is concentrated: AAPL and MSFT account for 49.68%, rising to 70.31% with META and TSLA. Continued AI-related earnings growth could sustain large-growth leadership; broader earnings participation could favor diversification; renewed inflation or financing pressure could penalize leverage. These are conditional investment scenarios, not forecasts validated by this project. Retain exposure without making the allocation depend on one style continuing to dominate.

Sources: 2026 tracking; shrinkage comparison; investment recommendation.

STAGE 4 | INVESTMENT-COMMITTEE MEMO (2 OF 2)

Proxy sensitivity and intellectual honesty

Roll's critique in our own results

Replacing French Mkt-RF with MSCI World USD net total return minus the same RF leaves the 30 stocks, 108 months and estimation methods fixed. MSCI returns include reinvested dividends after assumed withholding taxes; the change introduces international coverage and a different return convention, so it is not a pure geographic experiment. Both remain proxies for the unobservable all-asset market portfolio.

Statistic	French proxy	MSCI World
GRS F(30,77)	1.1465	1.4117
GRS p-value	0.3099	0.1154
Joint rejection at 5%	No	No
Individual significant alphas*	MSFT, WMT	AAPL, MSFT, WMT

*Two-sided p < 0.05, unadjusted for multiple testing; the lists agree under OLS and HAC(3).

Mean absolute changes are **0.0951 percentage points per month in alpha** and **0.0808 in beta**. TSLA has the largest absolute alpha change (+0.3702 pp/month); SCCO has the largest absolute beta change (+0.1915). AAPL's alpha rises from 1.0820% to 1.2617% per month, with its OLS p-value moving from 0.0754 to 0.0455. Individual estimates and significance labels are proxy-sensitive even though the 5% joint decision is stable.

Roll (1977) cautions that empirical CAPM tests depend on the market proxy's mean-variance properties. These results cannot cleanly separate failure of the model from inadequacy of the chosen proxy. Neither joint non-rejection establishes the CAPM nor does the smaller MSCI p-value prove that one proxy is correct. A switch from insignificant to significant is also not a formal test that the two alpha estimates differ. Full stock-level changes and the SML comparison are documented in Appendix F.

The result we trust least

We trust least the interpretation of the tangency portfolio's in-sample Sharpe of 2.197 as future achievable performance. Weights were selected and evaluated on the same historical observations, allowing the optimizer to exploit incidental sample patterns. The rise in gross exposure from 5.389 to 8.812 times capital when the window shrinks from 108 to 84 months demonstrates sensitivity. The strong 2026 tracking return is useful supplementary evidence, but retrospective selection and incomplete cost and margin treatment prevent it from resolving this concern. Our skepticism concerns generalization, not the arithmetic: a credible implementation case requires strategy rules fixed in advance, realistic constraints and further out-of-sample evaluation with comprehensive costs.

Sources: alpha/beta comparison; GRS comparison; limitations assessment.

Committee conclusion: the analysis supports cautious, constrained active allocation and transparent model sensitivity, rather than an unrestricted optimizer or a claim that the textbook has been conclusively validated.

APPENDIX A | CONTRIBUTIONS

Contribution statement and AI acknowledgment

The responsibilities below reflect the team's stated division of work. Each member should complete the verification / review column and sign before submission.

Team member	Work performed	Verification / review	Signature
Zhang Boyi	Data and universe: selection of 30 capitalization-based representatives. Stage 1: moments, static portfolio weights and subsample sensitivity.		
He Yuyang	Stage 2: CAPM regressions, SML and GRS test; integration into the project entry point.		
Leng Bojun	Stage 3: CAPM versus FF3, SMB/HML loading tests and alpha comparison; integration into the project entry point.		
Wang Mingyuan	Stage 4: investment recommendation, market-proxy comparison and limitations.		

AI acknowledgment

OpenAI Codex assisted with data auditing, universe-selection implementation, comparison figures and English documentation. The authors checked the results and remain responsible for all analysis and claims.

The acknowledgment above is reproduced from docs/contributions.md. It is the repository's suggested wording; members should confirm that the stated review reflects their actual checks before signing.

Source: docs/contributions.md.

Appendix navigation

Appendix	Supporting material
B	References, data provenance and reproducibility
C1-C4	Stage 1 full moments and weights; extreme observations; covariance and sensitivity figures
D	Stage 2 OLS and HAC alpha inference
E	Stage 3 factor loadings, alpha inference and loading tests
F1-F2	Stage 4 complete proxy comparison and alternative SML
G1-G3	Stage-by-stage artifact index, including large machine-readable matrices and time series

APPENDIX B | SOURCES AND REPRODUCTION

References and reproducibility boundary

Research references

Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3-56.

Gibbons, M. R., Ross, S. A., and Shanken, J. (1989). A test of the efficiency of a given portfolio. Econometrica, 57(5), 1121-1152.

Roll, R. (1977). A critique of the asset pricing theory's tests, Part I: On past and potential testability of the theory. Journal of Financial Economics, 4(2), 129-176.

Data and project sources

Kenneth French Data Library: monthly F-F Research Data Factors (Mkt-RF, SMB, HML and RF), via the instructor-supplied cached file. MSCI: World Standard large/mid-cap USD net-return index, code 990100, historical month-end levels. Massive and SEC EDGAR: upstream prices, corporate actions and issuer/text information. TBIC processed panels and historical IWV holdings underpin the retrospective security sample. The course handout is FE5108 Midterm Group Project, 2026 Teaching Edition.

Report source snapshot: main commit **d95edd3c3c7c4fbdc5a3a5955423e9961ced5e95**.
Supplementary research snapshot: **4eacac41c3fca3c02262c46ed0a1b5a904e087d0**.
Main project snapshot | Research tracking snapshot

Reproduce the analytical outputs

```
python3 -m pip install -r requirements.txt
python3 run_all.py --all
```

The unified entry point invokes the research/universe checkpoint and Stages 1-4 in order using versioned cached inputs. It does not reconstruct upstream text models, regenerate the separate research-branch experiments, or assemble the written report. Stage 4 recommendation and limitations documents are authored text.

To rebuild monthly returns from daily panels, the local tbic_returns directory must be supplied; it is excluded from Git. The --daily-root and --monthly options enable that step. Provider receipts, security continuity and full raw-data ingestion cannot be independently reconstructed from the repository alone. Accordingly, the delivered code is reproducible from the documented cached checkpoint, not from the complete provider data lake.

Package	Repository verified version
Python	3.12.1
NumPy	2.4.6
pandas	3.0.3
PyArrow	24.0.0
SciPy	1.18.0
scikit-learn	1.9.0
Matplotlib	3.10.9
statsmodels	0.14.6

Version table transcribed from the repository README; it describes its recorded verification environment, not the PDF-rendering environment. This report reads the pinned saved outputs and redraws figures from their tables; it does not claim a new independent raw-data audit or a fresh execution of the entire latest pipeline.

Numerical table cells are rounded for presentation; source CSVs retain precision. All clickable report links are confined to the project's GitHub repository. Large matrices, fitted-value panels and residual panels are indexed in Appendix G rather than printed cell by cell.

APPENDIX C1 | STAGE 1 DATA

Full means and portfolio weights

Table C1. Means are monthly percentages; weights are percentages of net asset value. Value weights are unchanged across sample windows. The full-sample means in the table correspond to 108 months.

Stock	Mean total	Mean excess	Value weight	Tangency 108m	Tangency 96m	Tangency 84m
SHW	1.554	1.362	0.52	-8.52	2.88	2.33
AAPL	2.486	2.294	26.22	11.53	6.58	12.69
ABBV	1.832	1.640	2.64	18.61	15.26	37.12
PLD	1.323	1.132	0.77	-1.58	-1.17	-12.85
TMO	1.570	1.378	1.42	18.63	23.18	32.29
SF	1.708	1.516	0.08	27.09	34.60	70.39
VZ	0.325	0.134	1.12	-32.64	-42.17	-55.52
META	2.192	2.000	10.86	-0.41	1.16	19.70
SCCO	2.255	2.063	0.77	18.46	20.26	14.64
MSFT	2.186	1.995	23.46	33.25	36.20	36.48
UPS	0.509	0.317	0.55	-27.97	-33.65	-14.57
WMT	1.764	1.573	5.80	31.42	20.63	0.99
PANW	2.563	2.371	0.84	8.96	13.64	0.58
RTX	0.947	0.755	1.61	-9.30	-8.53	-1.03
AFL	1.467	1.276	0.38	27.22	37.56	65.44
ACN	1.141	0.949	1.08	-42.96	-51.10	-54.59
V	1.625	1.433	4.41	41.36	39.01	39.35
EXEL	1.683	1.491	0.08	-1.57	-2.52	11.80
KO	0.860	0.669	1.96	41.24	57.94	78.95
UNH	1.111	0.920	1.95	2.20	-2.91	-14.02
WMB	1.387	1.195	0.48	19.20	21.24	44.09
MO	0.668	0.476	0.63	-4.35	-1.93	-12.22
AGNC	0.902	0.710	0.08	-21.67	-29.41	-77.98
CME	1.238	1.047	0.64	5.40	2.46	-35.39
SRE	0.953	0.762	0.38	-25.01	-23.24	-36.40
MAR	1.770	1.578	0.54	-16.96	-33.62	-32.69
CMG	1.971	1.780	0.32	0.56	11.67	-7.42
TSLA	4.857	4.665	9.76	7.54	11.80	16.51
SLB	0.312	0.121	0.37	-26.50	-23.03	-35.94
DHI	2.202	2.011	0.27	6.76	-2.81	7.28

All weight columns sum to 100% before rounding, but gross exposure is the sum of absolute weights. The value benchmark is company-capitalization weighted within this selected basket; it is neither free-float weighting nor a time-varying market index. Negative tangency weights represent short positions.

Sources: portfolio_weights.csv and means_108m.csv. Separate 96- and 84-month mean vectors and covariance matrices are listed in Appendix G1.

APPENDIX C2 | STAGE 1 DATA

Dispersion, extremes and capitalization

Table C2. Annualized mean = 12 × monthly arithmetic mean; annualized volatility = √12 × monthly standard deviation. Min/max returns are individual monthly observations. Market capitalization is the fixed December 31, 2025 vendor snapshot in USD billions.

Stock	Ann. mean (%)	Ann. vol. (%)	Min month (%)	Max month (%)	Cap (USD bn)
SHW	18.64	26.06	-18.64	17.55	80.32
AAPL	29.83	27.77	-18.12	21.66	4017.10
ABBV	21.98	25.86	-18.29	22.89	403.83
PLD	15.88	25.24	-21.63	17.78	118.58
TMO	18.84	23.78	-13.79	18.03	217.70
SF	20.49	29.35	-24.17	19.19	12.75
VZ	3.90	17.85	-11.71	14.25	171.74
META	26.30	36.39	-32.63	27.16	1663.78
SCCO	27.06	34.66	-19.38	31.73	117.52
MSFT	26.23	20.42	-10.93	17.63	3594.45
UPS	6.10	28.49	-17.29	28.40	84.15
WMT	21.17	19.06	-15.61	12.87	887.96
PANW	30.75	35.72	-25.82	32.88	128.39
RTX	11.36	28.18	-31.29	33.01	245.90
AFL	17.61	22.00	-20.09	30.24	57.79
ACN	13.69	23.77	-14.49	15.99	165.12
V	19.50	20.08	-11.65	16.61	676.18
EXEL	20.19	41.63	-21.94	46.43	11.75
KO	10.33	16.38	-16.57	12.89	300.73
UNH	13.34	26.89	-26.62	24.17	299.03
WMB	16.64	28.11	-23.49	36.89	73.41
MO	8.02	23.24	-21.28	19.38	96.79
AGNC	10.82	28.43	-37.06	23.03	11.50
CME	14.86	18.47	-12.67	16.13	98.47
SRE	11.44	19.99	-18.32	17.41	57.63
MAR	21.23	33.17	-39.67	36.60	83.25
CMG	23.66	34.90	-23.63	34.25	48.92
TSLA	58.28	65.70	-36.73	74.15	1495.69
SLB	3.75	43.20	-50.20	44.93	57.34
DHI	26.43	36.72	-36.17	38.88	41.93

The snapshot reconciles internally to raw close times vendor weighted shares, but neither share-count accuracy nor historical availability is independently established by that arithmetic. It differs from the December 2024 capitalization used to select the universe.

Sources: summary statistics; capitalization audit.

APPENDIX C3 | STAGE 1 FIGURE

Sensitivity to the estimation window

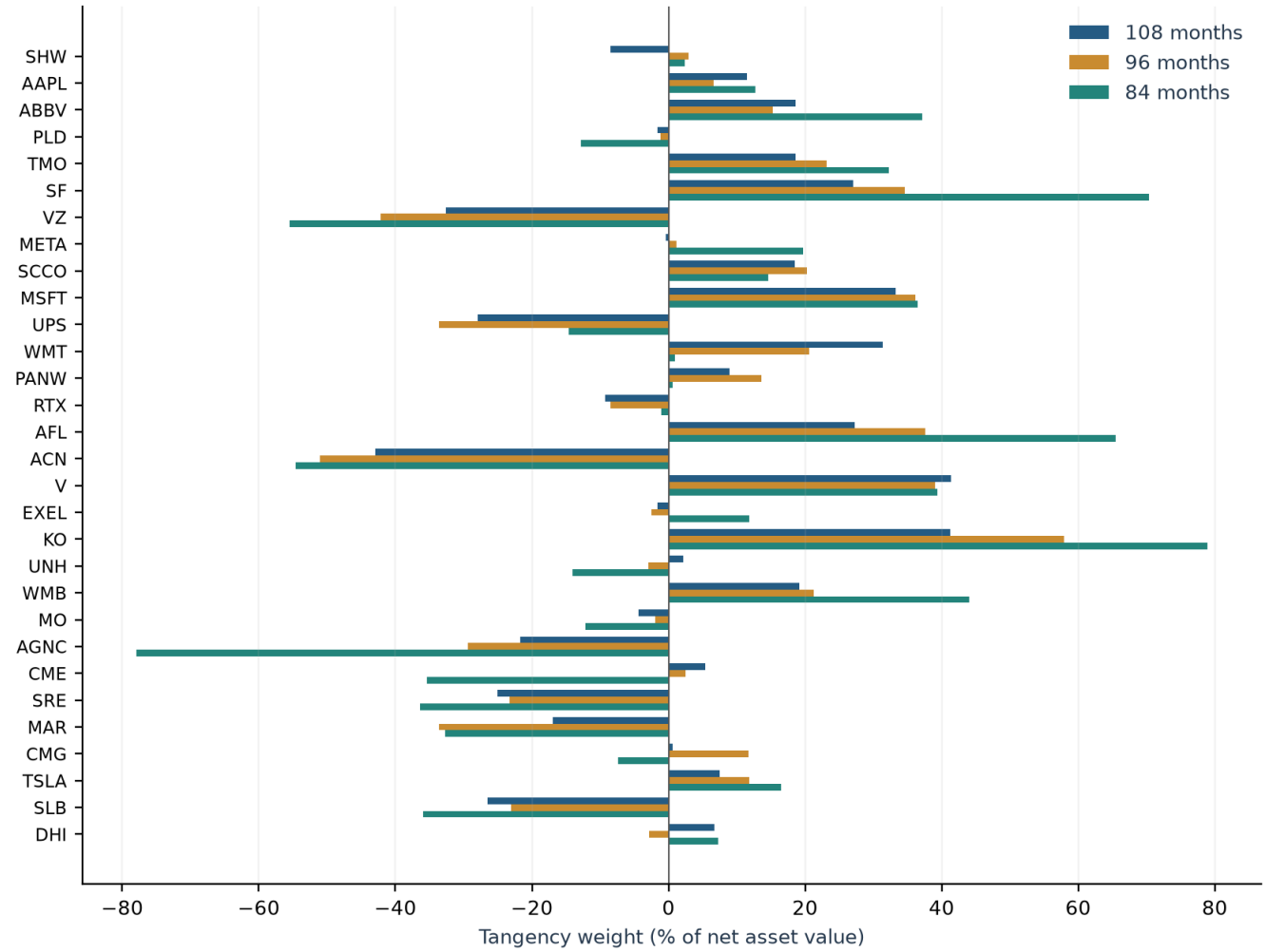


Figure C1. Tangency weights across estimation windows. Takeaway: shortening the sample changes individual allocations substantially and raises gross exposure, even with the same stocks and endpoint. This combines estimation uncertainty and potential regime differences; it does not isolate either cause.

Window	Same sign as 108m	Mean |Δw| (pp)	Max |Δw| (pp)	Gross exposure
96 months	26/30	5.85	16.70	6.122×
84 months	24/30	17.08	56.31	8.812×

The 96-month sample covers January 2018-December 2025; the 84-month sample covers January 2019-December 2025. The latter is only a robustness exercise and does not replace the qualifying 108-month baseline.

Sources: weights; stability measures. Repository figure: subsample_weights.png.

APPENDIX C4 | STAGE 1 COVARIANCE

Dependence structure of the selected assets

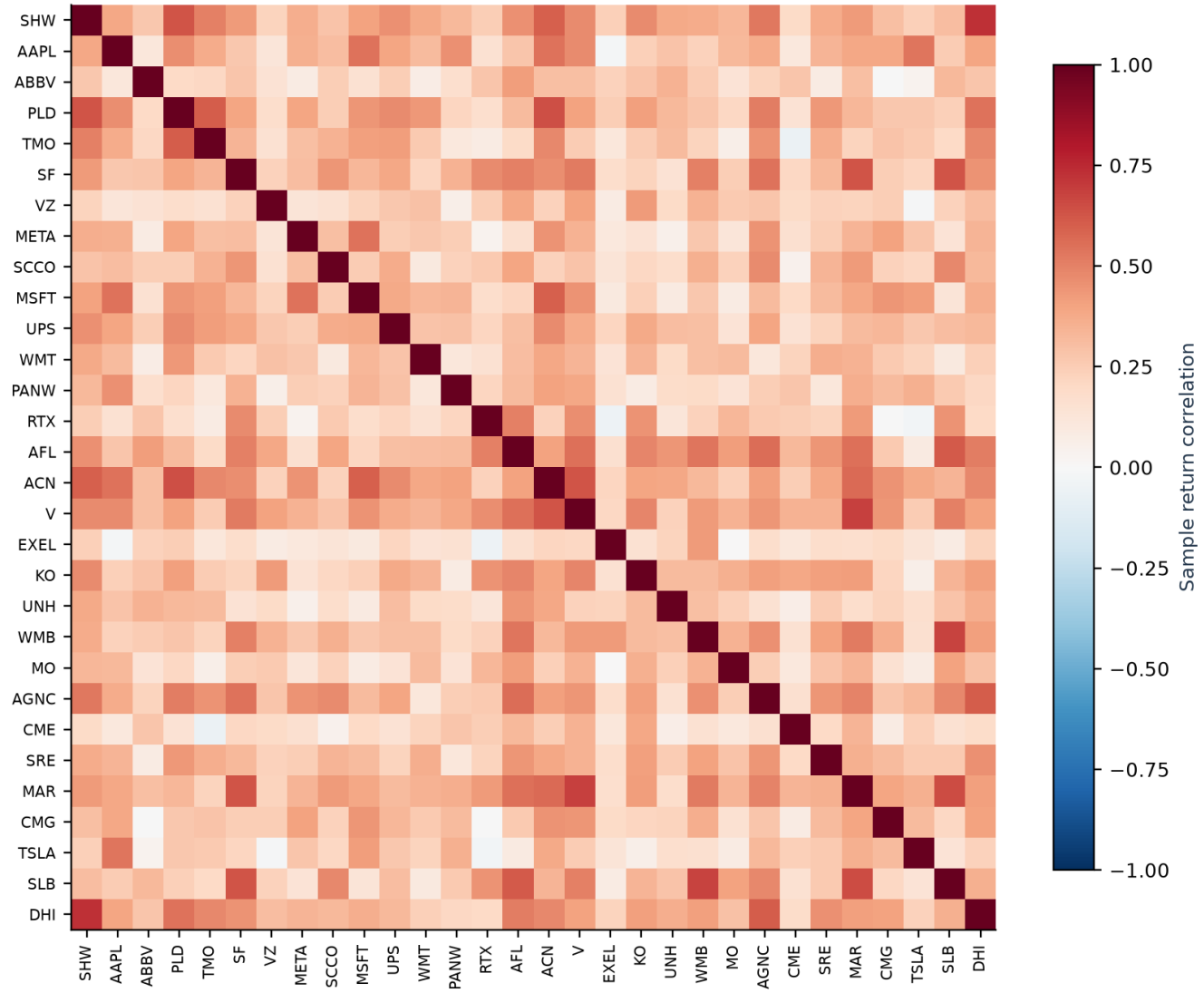


Figure C2. Correlations derived from the saved sample covariance. Takeaway: one representative per text class does not remove return co-movement; the optimizer still depends on estimated off-diagonal covariances. This display standardizes the covariance matrix for readability and is not a replacement estimator.

Each correlation is $\Sigma_{ij}/\sqrt{(\Sigma_{ii}\Sigma_{jj})}$. The original covariance uses monthly total-return decimals and divisor T - 1. Its condition number is 138.65 in the 108-month sample, rising to 166.32 and 261.18 for 96 and 84 months. This is compatible with sensitivity to estimated inputs; it is not by itself a formal diagnosis of a faulty matrix.

Source: artifacts/stage1/covariance_monthly_108m.csv.

The full 30 × 30 matrices for all three windows are available as machine-readable CSVs in Appendix G1's index.

APPENDIX D | STAGE 2 INFERENCE

Individual alpha inference: OLS and HAC

Table D1. French market proxy; alpha in percent per month. OLS and HAC(3) p-values are two-sided and unadjusted for multiple comparisons. A nominal 5% threshold is used for significance lists; |t| > 2 is retained as the course's descriptive screen.

Stock	Alpha	OLS t	OLS p	HAC(3) t	HAC(3) p
SHW	0.191	0.347	0.7293	0.414	0.6796
AAPL	1.082	1.796	0.0754	1.891	0.0614
ABBV	1.065	1.522	0.1310	1.843	0.0681
PLD	0.039	0.071	0.9432	0.079	0.9372
TMO	0.492	0.867	0.3877	0.944	0.3471
SF	0.130	0.220	0.8264	0.229	0.8190
VZ	-0.229	-0.472	0.6376	-0.527	0.5994
META	0.699	0.798	0.4264	0.789	0.4317
SCCO	0.890	1.042	0.2997	1.045	0.2982
MSFT	1.032	2.495	0.0141	2.342	0.0211
UPS	-0.775	-1.157	0.2500	-1.226	0.2229
WMT	1.008	2.070	0.0409	2.342	0.0211
PANW	1.159	1.316	0.1909	1.618	0.1086
RTX	0.018	0.024	0.9811	0.024	0.9812
AFL	0.430	0.837	0.4047	0.761	0.4486
ACN	-0.254	-0.572	0.5689	-0.538	0.5915
V	0.515	1.232	0.2206	1.283	0.2024
EXEL	0.787	0.682	0.4965	0.827	0.4103
KO	0.201	0.474	0.6367	0.438	0.6622
UNH	0.330	0.451	0.6526	0.529	0.5979
WMB	0.084	0.130	0.8972	0.124	0.9014
MO	-0.058	-0.093	0.9264	-0.098	0.9224
AGNC	-0.558	-0.925	0.3569	-0.975	0.3316
CME	0.698	1.378	0.1712	1.324	0.1885
SRE	0.065	0.133	0.8945	0.126	0.9002
MAR	-0.009	-0.013	0.9893	-0.013	0.9896
CMG	0.567	0.665	0.5076	0.628	0.5313
TSLA	2.491	1.524	0.1305	1.559	0.1221
SLB	-1.550	-1.536	0.1275	-1.343	0.1823
DHI	0.408	0.514	0.6085	0.615	0.5397

MSFT and WMT have $p < 0.05$ under both standard-error conventions. The GRS test still uses classical residual covariance and its own joint F reference, rather than combining individual HAC p-values.

Source: artifacts/stage2/capm_results.csv.

APPENDIX E | STAGE 3 INFERENCE

Full FF3 estimates and factor-loading tests

Table E1. Alpha in percent per month; factor loadings are unitless. The last column tests SMB and HML loadings jointly equal to zero for each stock. It is not a joint-alpha test.

Stock	FF3 alpha	OLS t	HAC p	Market beta	SMB beta	HML beta	Loading p
SHW	0.237	0.42	0.6428	1.087	-0.005	0.208	0.326
AAPL	0.909	1.55	0.0945	1.155	-0.151	-0.456	0.0063
ABBV	1.099	1.58	0.0630	0.560	-0.142	0.432	0.0448
PLD	0.062	0.11	0.9008	0.999	0.083	-0.065	0.84
TMO	0.438	0.76	0.3943	0.828	-0.023	-0.187	0.418
SF	0.461	0.89	0.3349	1.217	0.356	0.740	7.6e-08
VZ	-0.444	-0.97	0.2548	0.451	-0.613	0.270	0.00028
META	0.331	0.39	0.6801	1.310	-0.537	-0.548	0.00736
SCCO	0.999	1.17	0.2420	1.089	-0.010	0.495	0.0685
MSFT	0.693	2.07	0.0454	0.984	-0.473	-0.545	2.94e-11
UPS	-0.897	-1.31	0.1391	1.063	-0.263	-0.013	0.568
WMT	0.737	1.56	0.0659	0.623	-0.528	-0.141	0.00455
PANW	1.244	1.41	0.1210	1.054	0.386	-0.391	0.121
RTX	0.054	0.08	0.9253	0.733	-0.272	0.695	0.000499
AFL	0.457	1.02	0.2819	0.835	-0.284	0.678	7.75e-08
ACN	-0.376	-0.84	0.4179	1.148	-0.160	-0.215	0.0841
V	0.336	0.81	0.3638	0.932	-0.428	0.063	0.0188
EXEL	0.777	0.66	0.4392	0.655	-0.004	-0.034	0.993
KO	-0.020	-0.05	0.9570	0.558	-0.668	0.351	2.51e-07
UNH	0.391	0.53	0.5272	0.546	0.004	0.258	0.373
WMB	0.161	0.28	0.7631	1.071	-0.225	0.781	2.49e-06
MO	-0.096	-0.16	0.8523	0.565	-0.379	0.582	0.000258
AGNC	-0.378	-0.66	0.4923	1.155	0.111	0.567	0.00047
CME	0.643	1.25	0.2163	0.355	-0.170	0.095	0.529
SRE	0.040	0.08	0.9310	0.677	-0.166	0.220	0.149
MAR	0.154	0.24	0.7826	1.459	0.062	0.588	0.00128
CMG	0.400	0.47	0.6494	1.145	-0.093	-0.547	0.0332
TSLA	2.732	1.73	0.0537	1.807	1.157	-1.230	0.00216
SLB	-0.997	-1.35	0.1702	1.485	0.311	1.797	6.57e-16
DHI	0.652	0.82	0.3625	1.425	0.328	0.417	0.0491

Only MSFT has FF3 alpha $p < 0.05$ under either OLS or HAC(3). Individual loading tests show substantial exposure heterogeneity. The full CSV also preserves CAPM/FF3 adjusted R^2 and individual loading p-values; those supplementary columns remain machine-readable.

Source: artifacts/stage3/all_30_model_results.csv.

APPENDIX F1 | STAGE 4 DATA

Full market-proxy comparison

Table F1. Alpha in percent per month; Δalpha in percentage points per month. Δ = MSCI World minus French. MSCI OLS p-values are unadjusted for multiplicity.

Stock	French alpha	MSCI alpha	Δalpha	French beta	MSCI beta	Δbeta	MSCI p
SHW	0.191	0.294	+0.102	1.087	1.182	+0.094	0.5897
AAPL	1.082	1.262	+0.180	1.125	1.142	+0.016	0.0455
ABBV	1.065	1.079	+0.014	0.535	0.621	+0.087	0.1202
PLD	0.039	0.144	+0.105	1.014	1.092	+0.078	0.7923
TMO	0.492	0.588	+0.096	0.823	0.874	+0.051	0.3021
SF	0.130	0.309	+0.179	1.287	1.335	+0.048	0.6130
VZ	-0.229	-0.234	-0.004	0.337	0.406	+0.069	0.6247
META	0.699	0.806	+0.107	1.208	1.321	+0.113	0.3536
SCCO	0.890	0.905	+0.015	1.089	1.281	+0.191	0.2721
MSFT	1.032	1.153	+0.121	0.894	0.931	+0.037	0.0076
UPS	-0.775	-0.672	+0.104	1.014	1.093	+0.079	0.3152
WMT	1.008	1.068	+0.060	0.524	0.558	+0.034	0.0301
PANW	1.159	1.326	+0.167	1.125	1.155	+0.031	0.1390
RTX	0.018	0.006	-0.012	0.685	0.829	+0.144	0.9934
AFL	0.430	0.439	+0.009	0.785	0.925	+0.140	0.3668
ACN	-0.254	-0.113	+0.141	1.117	1.174	+0.057	0.8042
V	0.515	0.577	+0.062	0.853	0.946	+0.094	0.1547
EXEL	0.787	0.898	+0.111	0.654	0.656	+0.002	0.4378
KO	0.201	0.175	-0.027	0.434	0.547	+0.112	0.6668
UNH	0.330	0.398	+0.068	0.548	0.577	+0.029	0.5862
WMB	0.084	0.203	+0.119	1.032	1.098	+0.066	0.7549
MO	-0.058	-0.047	+0.011	0.496	0.579	+0.083	0.9389
AGNC	-0.558	-0.501	+0.057	1.178	1.339	+0.162	0.3811
CME	0.698	0.726	+0.028	0.324	0.355	+0.031	0.1522
SRE	0.065	0.126	+0.061	0.647	0.703	+0.056	0.7951
MAR	-0.009	0.118	+0.127	1.473	1.615	+0.141	0.8549
CMG	0.567	0.711	+0.144	1.126	1.182	+0.056	0.4082
TSLA	2.491	2.861	+0.370	2.018	1.995	-0.024	0.0890
SLB	-1.550	-1.430	+0.120	1.551	1.715	+0.164	0.1508
DHI	0.408	0.544	+0.136	1.488	1.622	+0.134	0.4878

The core inference appears on p. 10. This appendix provides the complete supporting table, rather than extending the two-page investment-committee memo. Significance changes are not formal tests of coefficient differences.

Source: artifacts/stage4/alpha_beta_comparison.csv.

APPENDIX F2 | STAGE 4 FIGURE

Alternative-proxy Security Market Line

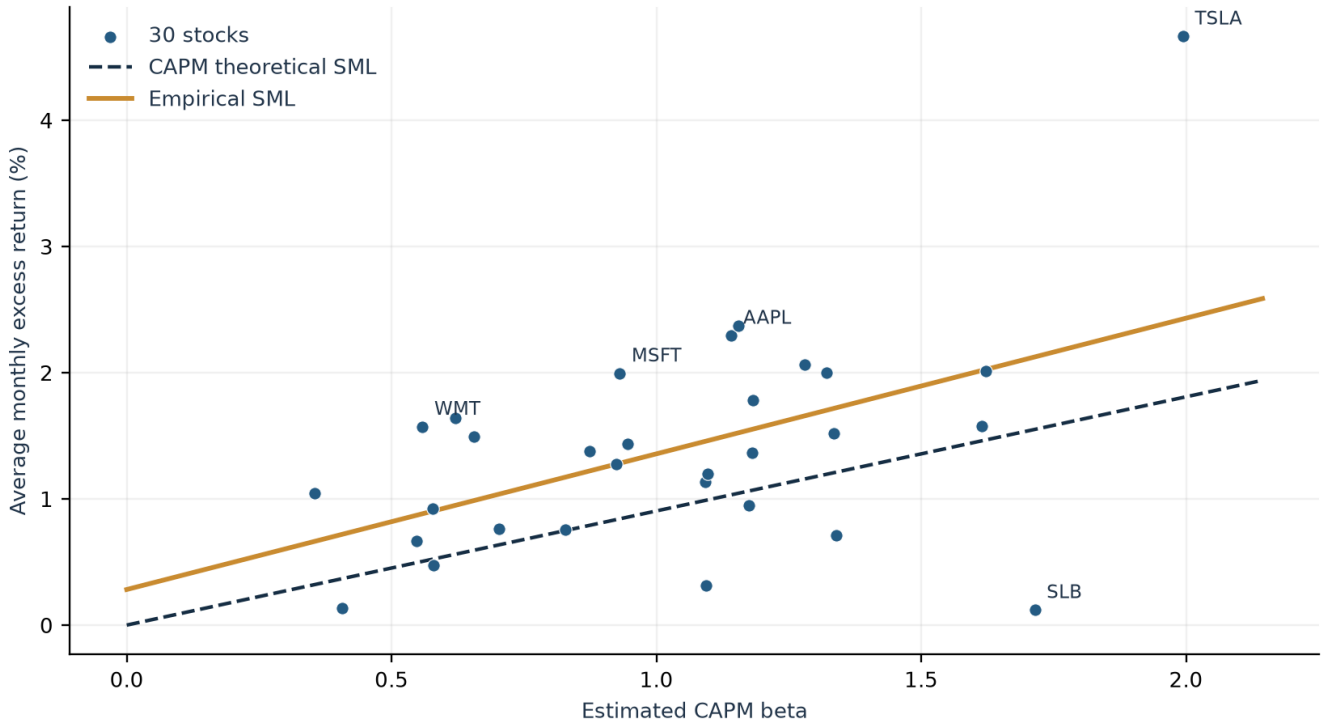


Figure F1. MSCI World theoretical and empirical SML. Takeaway: the positive fitted intercept persists under the replacement proxy, while the theoretical excess-return line still passes through zero. Both estimated slope and market premium differ from the French specification.

Quantity	French proxy	MSCI World
Theoretical intercept	0.0000%	0.0000%
Empirical intercept	0.2108%	0.2819%
Theoretical slope	1.0771%	0.9044%
Empirical slope	1.2396%	1.0748%
Cross-sectional R^2	0.3164	0.2494

Intercepts and slopes are expressed in monthly percentage units; slopes apply per unit beta. Cross-sectional fit declines with the alternative proxy in this sample, but that comparison does not identify the true market portfolio. Neither table is a formal test that the two fitted SMLs differ.

Sources: French SML summary; MSCI SML summary; MSCI plotting data.

APPENDIX G1 | OUTPUT CATALOG

Stages 1 and 2: artifact index

Every listed file is stored beneath artifacts/stage1, stage2, stage3 or stage4 in the pinned main snapshot. Clickable file references point only to this project's GitHub repository. Large time-series panels and covariance matrices are indexed rather than printed as hundreds of cells.

Stage	Files	Role / report location
Stage 1	README.md market_cap_audit.md	Method, full weights and capitalization limitations
Stage 1	summary_statistics.csv means_108m.csv means_96m.csv means_84m.csv	Universe moments; main Table 1 and Appendices C1-C2
Stage 1	portfolio_weights.csv weights_108m.csv weights_96m.csv weights_84m.csv	Tangency and value allocations; Figure 1 and Table C1
Stage 1	covariance_monthly_108m.csv covariance_monthly_96m.csv covariance_monthly_84m.csv	Complete 30 × 30 covariance matrices; Figure C2 summarizes the full sample
Stage 1	portfolio_diagnostics.csv subsample_stability.csv	Sharpe, gross/net exposure and window sensitivity
Stage 1	weights_comparison.png subsample_weights.png	Repository counterparts of report Figures 1 and C1
Stage 1	verification.json	Input hashes and calculation checks
Stage 2	capm_results.csv capm_summary.csv	All stock regressions; Table 2 and Appendix D
Stage 2	capm_input_monthly.csv data_audit_examples.csv	Aligned 108-month input panel and accounting examples
Stage 2	capm_fitted_values.csv capm_residuals.csv	Complete 108 × 30 fitted-value and residual panels
Stage 2	sml_points.csv sml_lines.csv sml_summary.csv sml_scatter.png	SML coordinates, coefficients and original chart; Figure 2
Stage 2	grs_results.csv grs_residual_covariance.csv	Joint test and complete 30 × 30 residual covariance
Stage 2	input_checks.json capm_checks.json sml_checks.json grs_checks.json	Data, regression, SML and GRS diagnostics

The report redraws the core figures from saved numeric outputs for consistent print sizing. It does not change the estimators or replace the original machine-readable outputs.

APPENDIX G2 | OUTPUT CATALOG

Stages 3 and 4: artifact index

Every listed file is stored beneath artifacts/stage1, stage2, stage3 or stage4 in the pinned main snapshot. Clickable file references point only to this project's GitHub repository. Large time-series panels and covariance matrices are indexed rather than printed as hundreds of cells.

Stage	Files	Role / report location
Stage 3	README.md AUDIT.md	Methods, reproduction and review notes
Stage 3	factor_definitions.md factor_interpretation.md	Required factor descriptions and surviving-alpha interpretation
Stage 3	alpha_comparison.csv alpha_comparison.png	All-stock before/after alphas; Figure 3
Stage 3	all_30_model_results.csv checks.json	Full estimates, loading tests and input checks; Appendix E
Stage 3	run_stage3.py	Compatibility wrapper; canonical calculation code is scripts/stage3.py
Stage 4	investment_recommendation.md limitations.md	Authored narrative sources, condensed into the two-page memo
Stage 4	alpha_beta_comparison.csv grs_comparison.csv comparison_summary.json comparison_report.md	French versus MSCI changes and readable report; Table F1
Stage 4	capm_results.csv capm_summary.csv	Full MSCI CAPM estimates and presentation summary
Stage 4	capm_input_monthly.csv data_audit_examples.csv capm_fitted_values.csv capm_residuals.csv	Alternative-market input panel, audit examples, fitted values and residuals
Stage 4	sml_points.csv sml_lines.csv sml_summary.csv sml_scatter.png	Alternative SML plotting data, coefficients and chart; Figure F1
Stage 4	grs_results.csv grs_residual_covariance.csv	MSCI joint-alpha test and residual covariance
Stage 4	input_checks.json capm_checks.json sml_checks.json grs_checks.json	Data and method diagnostics for the alternative proxy

The report redraws the core figures from saved numeric outputs for consistent print sizing. It does not change the estimators or replace the original machine-readable outputs.

APPENDIX G3 | INPUTS AND COVERAGE

Evidence map and supplementary research

Source group	Location	Purpose
Data and selection	docs/01-data.md docs/03-universe-selection.md	Security identities, coverage rules, selection timing and upstream processing
Monthly inputs	artifacts/monthly/README.md	Stock total/excess returns, RF and MSCI definitions
Course-stage scripts	run_all.py scripts/stage1.py scripts/stage2.py scripts/stage3.py scripts/stage4.py	Unified analytical reproduction from cached inputs
Contribution record	docs/contributions.md	Contribution statement and AI acknowledgment (Appendix A)
Supplementary tracking	research_2026/README.md	Retrospective 2026 tracking; used only to support the Stage 4 recommendation
Covariance shrinkage	research_2026/ledoit_wolf/README.md	Same mean estimates, alternative covariance; brief comparison in Table 3

Required-exhibit coverage

Handout requirement	Report location
One-page executive summary	Page 1
1a: universe summary	Table 1, p. 3; Appendices C1-C2
1b: tangency and value weights in one chart	Figure 1, p. 4
1c: estimation-error / subsample discussion	Page 4 and Figure C1
2a: beta, alpha, t(alpha), R^2	Table 2, p. 5
2b and 2d: empirical/theoretical SML and interpretation	Figure 2 and SML table, p. 6
2c: joint alpha assessment	GRS test and interpretation, p. 6
3a: two additional factor definitions	SMB and HML paragraphs, p. 7
3b and 3c: alpha comparison and verdict	Pages 7-8, Figure 3
Stage 4: three investment-committee questions	Pages 9-10 only; supporting exhibits in Appendix F
Contribution statement and AI disclosure	Appendix A; verification and signatures left blank

Separate exploratory text-distance allocation experiments are not a required course stage and are not used to claim classification skill in this report. The 2026 evidence has an October 6 endpoint, not a full-year endpoint.